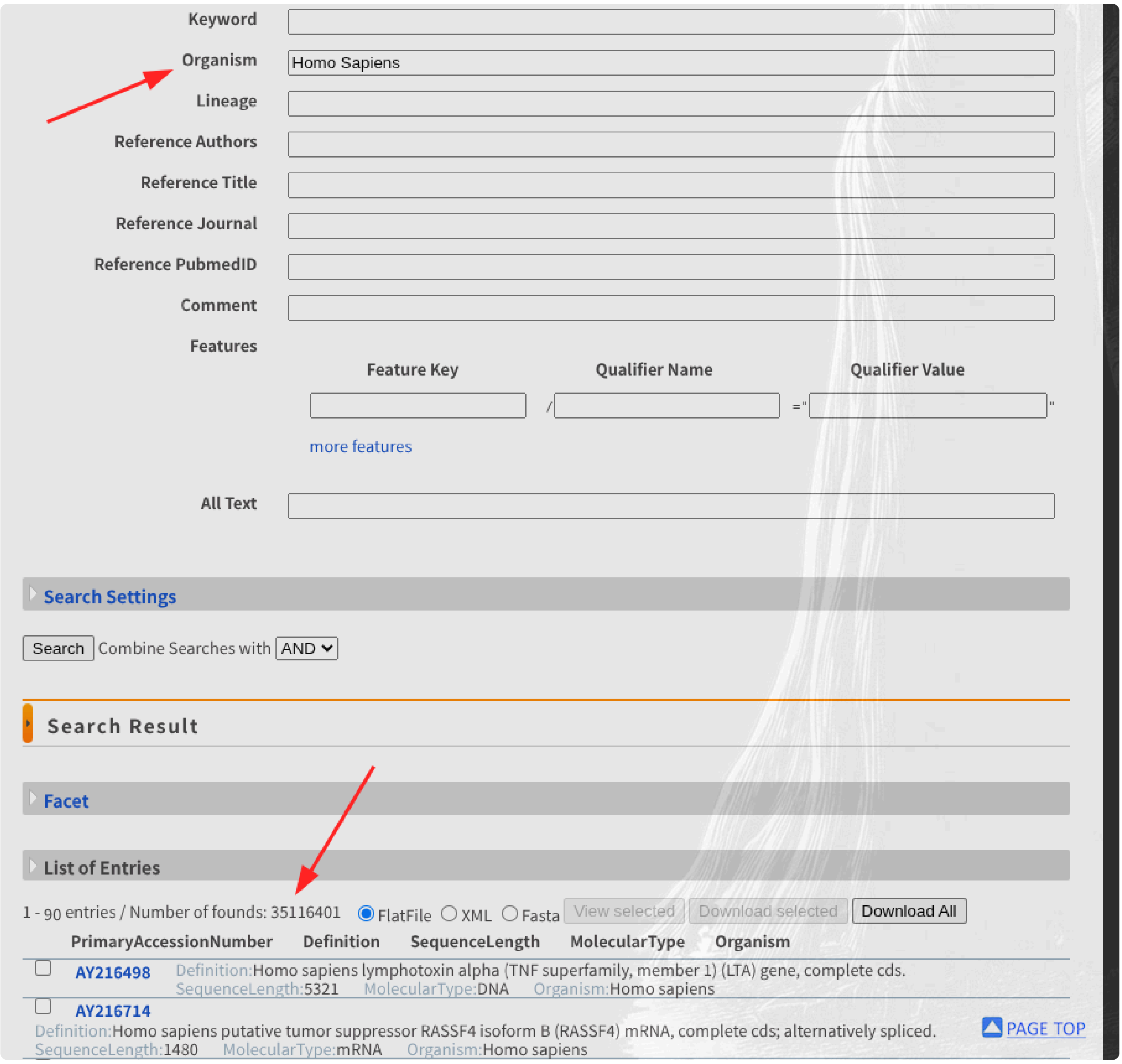
Solution - 2

Q1

DDJB

this link opens the DDJB advanced search for Homo sapiens



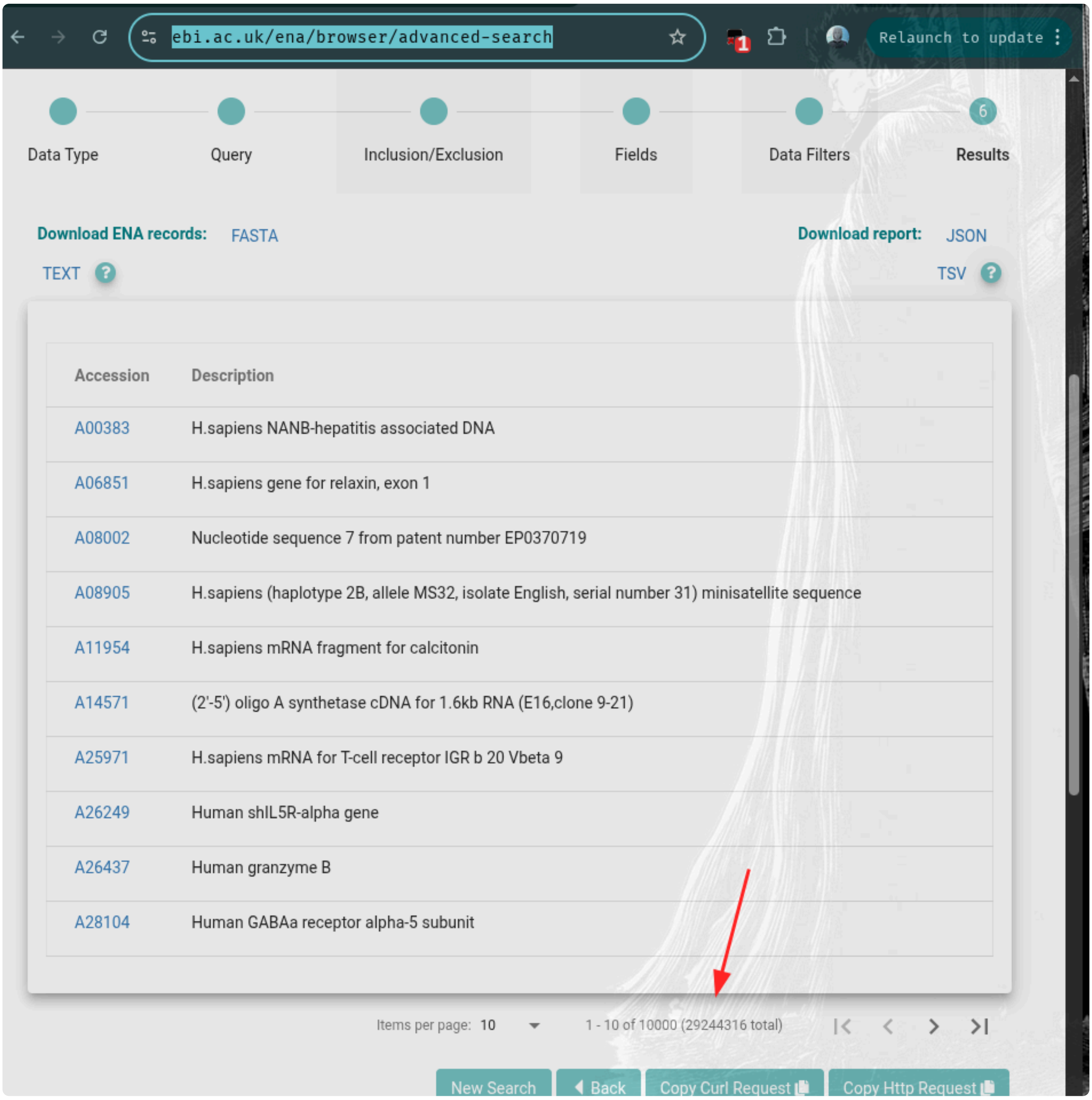
so DDBJ = **35,116,401**

genbank

in the nucleotide database in genbank it shows **29,841,998**

EMBL

this link



so **29,244,316**

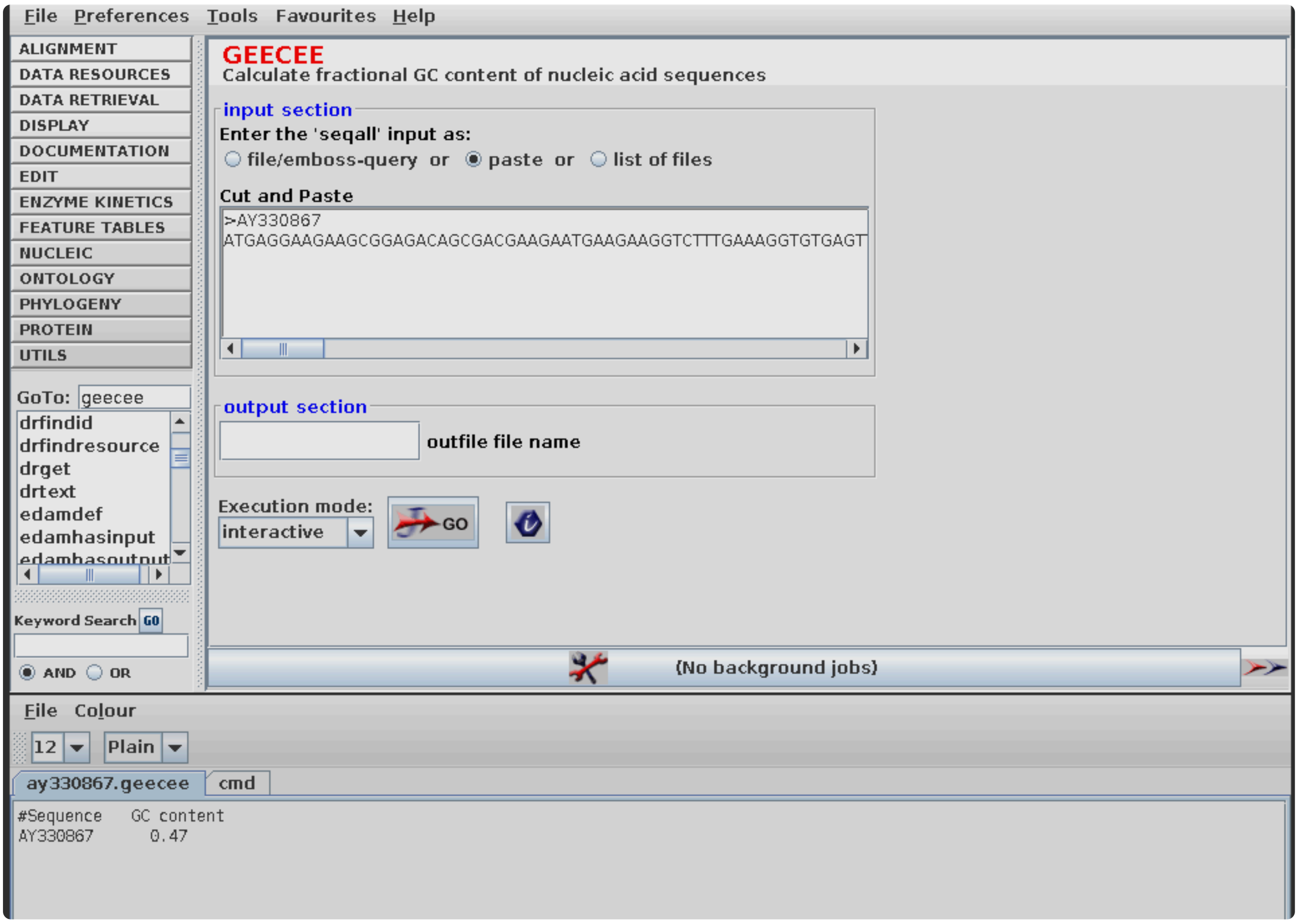
finally
DDBJ = 35,116,401
GenBank = 29,841,998
EMBL = 29,224,316

Q.2

link
this is the sequence taken from genbank

```
ATGAGGAAGAAGCGGAGACAGCGACGAAGAATGAAGAAGGTCTTTGAAAGGTGTGA
GTTGGCCAGAACTC
TGAAAAGATTGGGAATGGATGGCTACAGGGGAATCAGCCTAGCAAACTGGATGTGT
TTGGCCAAATGGGA
GAGTGGTTACAACACACGAGCTACAAACTACAATGCTGGAGACAGAAGCACTGATT
ATGGGATATTTCAG
ATCAATAGCCGCTACTGGTGTAATGATGGCAAAACCCCAGGAGCAGTTAATGCCTG
TCATTTATCCTGCA
GTGCTTTGCTGCAAGATAACATCGCTGATGCTGTAGCTTGTGCAAAGAGGGTTGTC
CGTGATCCACAAGG
CATTAGAGCATGGGTGGCATGGAGAAATCGTTGTCAAAACAGAGGTGTCCGTCAGT
ATGTTCAAGGTTGT
GGAGTGTAATGA
```

using emboss we can figure out the GC content



so it is 47%

Q.3

- https://www.ncbi.nlm.nih.gov/nuccore/33088589

- https://getentry.ddbj.nig.ac.jp/getentry/na/AY330867/?format=flatfile&filetype=html&trace=true&show_suppressed=false&limit=10
- https://www.ebi.ac.uk/ena/browser/view/AY330867?show=orcid-links

all databases give the sewuence, translation and other useful info
all three are part of the INSDC and sync daily, so the core biological sequences and accession numbers are identical across all of them
(taken from ddbj)

```
LOCUS       AY330867                 432 bp    mRNA    linear   SYN 29-JUL-2004
DEFINITION  Synthetic construct human lysozyme mRNA, complete cds.
ACCESSION   AY330867
VERSION     AY330867.1
KEYWORDS    .
SOURCE      synthetic construct
  ORGANISM  synthetic construct
            other sequences; artificial sequences.
REFERENCE   1  (bases 1 to 432)
  AUTHORS   Xie,X.D., Chen,Z.H. and Zhou,C.S.
  TITLE     Direct Submission
  JOURNAL   Submitted (26-JUN-2003) Postdoctoral Scientific Research Station of
            Gansu Yasheng Industrial (Group) Co., Ltd., Zhangye Road, Lanzhou,
            Gansu 730030, P.R.China
FEATURES             Location/Qualifiers
     source          1..432
                     /organism="synthetic construct"
                     /mol_type="mRNA"
                     /db_xref="taxon:32630"
     CDS             1..429
                     /codon_start=1
                     /transl_table=11
                     /product="human lysozyme"
                     /protein_id="AAP93336.1"
                     /translation="MRKKRRQRRRMKKVFERCELARTLKRLGMDGYRGISLANWMCLA
                     KWESGYNTRATNYNAGDRSTDYGIFQINSRYWCNDGKTPGAVNACHLSCSALLQDNIA
                     DAVACAKRVVRDPQGIRAWVAWRNRCQNRGVRQYVQGCGV"
BASE COUNT          131 a           72 c          129 g          100 t
ORIGIN
        1 atgaggaaga agcggagaca gcgacgaaga atgaagaagg tctttgaaag gtgtgagttg
       61 gccagaactc tgaaaagatt gggaatggat ggctacaggg gaatcagcct agcaaactgg
      121 atgtgtttgg ccaaatggga gagtggttac aacacacgag ctacaaacta caatgctgga
      181 gacagaagca ctgattatgg gatatttcag atcaatagcc gctactggtg taatgatggc
      241 aaaaccccag gagcagttaa tgcctgtcat ttatcctgca gtgctttgct gcaagataac
      301 atcgctgatg ctgtagcttg tgcaaagagg gttgtccgtg atccacaagg cattagagca
      361 tgggtggcat ggagaaatcg ttgtcaaaac agaggtgtcc gtcagtatgt tcaaggttgt
      421 ggagtgtaat ga
//
```

but they have some differences like

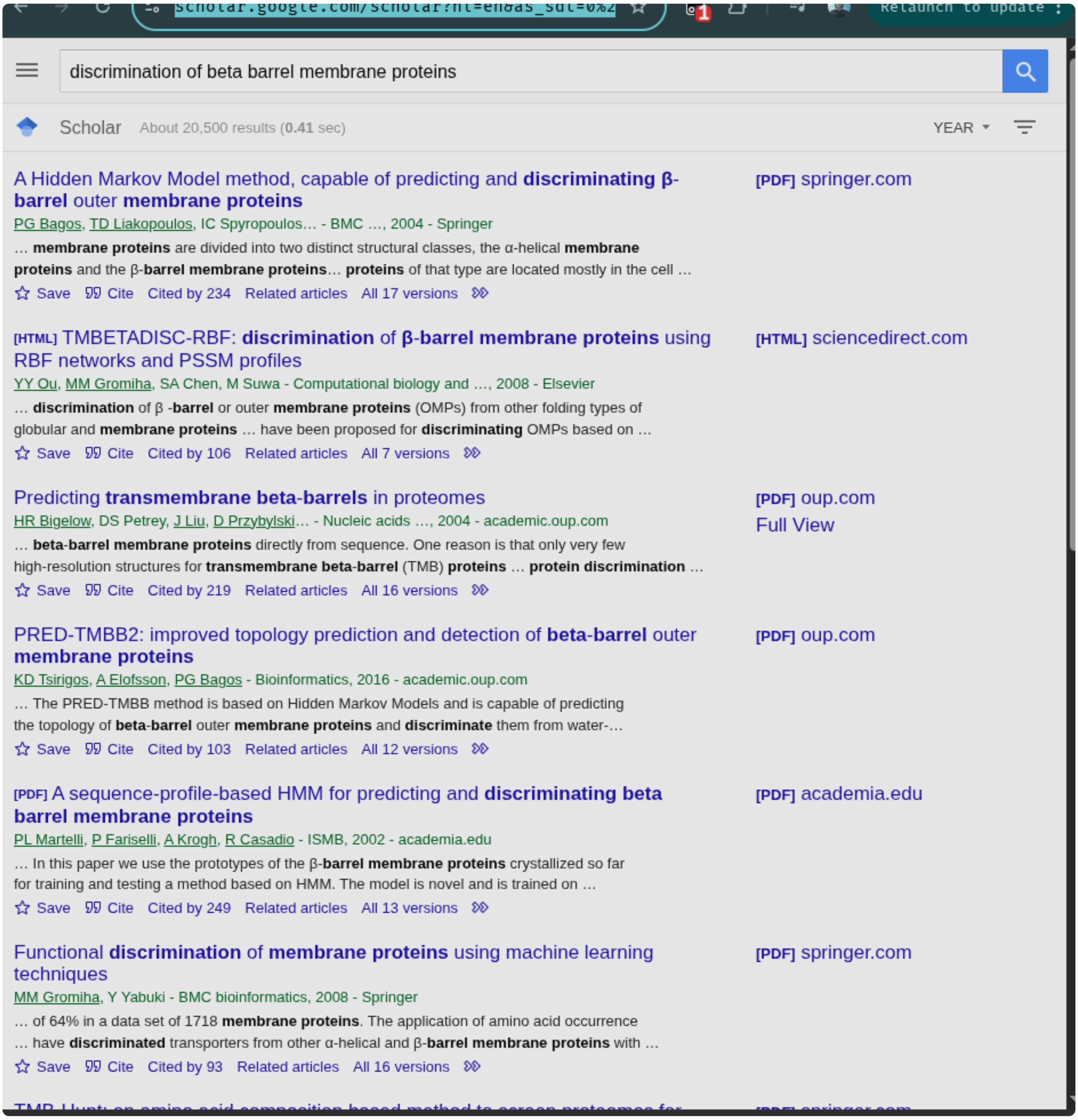
- genbank and ddbj use full words like locus and origin for headers but embl uses two letter codes like id and sq
- ddbj clearly shows the exact base count right before the sequence but genbank skips it completely

- embl web view gives extra stuff like orcid data claims and cross references to other databases
- genbank and ddbj put the sequence numbers on the left side but embl puts them on the far right
- **genbank** it is tied to NCBI, meaning it natively cross-links to PubMed for literature, BLAST for alignments, and RefSeq for curated genomes
- **embl** it is built into the european framework, directly linking nucleotide sequences to Ensembl (genome browsers) and UniProt (protein databases)
- **ddbj** it serves the asian research node and links out to specific functional databases like KEGG for metabolic pathway analysis

the raw sequence data is exactly the same; the real difference is just the web interface and the downstream analytical tools each database connects us to

Q.4

papers link



some examples:

Liu, Q., Zhu, Y., Wang, B., & Li, Y. (2003). Identification of β-barrel membrane proteins based on amino acid composition properties and predicted secondary structure. *Computational Biology and Chemistry*, *27*, 355–361. https://doi.org/10.1016/s1476-9271(02)00085-3 Cited by: 52

Ou, Y.-Y., Gromiha, M. M., Chen, S.-A., & Suwa, M. (2008). TMBETADISC-RBF: Discrimination of -barrel membrane proteins using RBF networks and PSSM profiles. *Computational Biology and*

Chemistry, *32*, 227–231.
https://doi.org/10.1016/j.compbiolchem.2008.03.002 Cited by: 106

Tsirigos, K. D., Elofsson, A., & Bagos, P. G. (2016). PRED-TMBB2: improved topology prediction and detection of beta-barrel outer membrane proteins. *Bioinformatics*, *32*, i665–i671.
https://doi.org/10.1093/bioinformatics/btw444 Cited by: 103

Q.5

Kihara D Google Scholar link
one paper published in nature journal example
paper link
they report the report the results from the first large-scale community-based critical assessment of protein function annotation (CAFA) experiment.

Q.6

paper link
"Mitochondrial beta-barrel proteins, an exclusive club?"
there are lots of related articles

Similar articles

Transport proteins (carriers) of mitochondria.

Wohlrab H.

IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.

PMID: 18816452 Review.

Systematic analysis of the twin cx(9)c protein family.

Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.

J Mol Biol. 2009 Oct 23;393(2):356-68. doi: 10.1016/j.jmb.2009.08.041. Epub 2009 Aug 21.

PMID: 19703468

Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.

Javadov S, Karmazyn M, Escobales N.

J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.

PMID: 19509316

Vitamin E protects against the mitochondrial damage caused by cyclosporin A in LLC-PK1 cells.

de Arriba G, de Hornedo JP, Rubio SR, Fernández MC, Martínez SB, Camarero MM, Cid TP.

Toxicol Appl Pharmacol. 2009 Sep 15;239(3):241-50. doi: 10.1016/j.taap.2009.05.028. Epub 2009 Jun 11.

PMID: 19523970

A CaPful of mechanisms regulating the mitochondrial permeability transition.

Di Lisa F, Bernardi P.

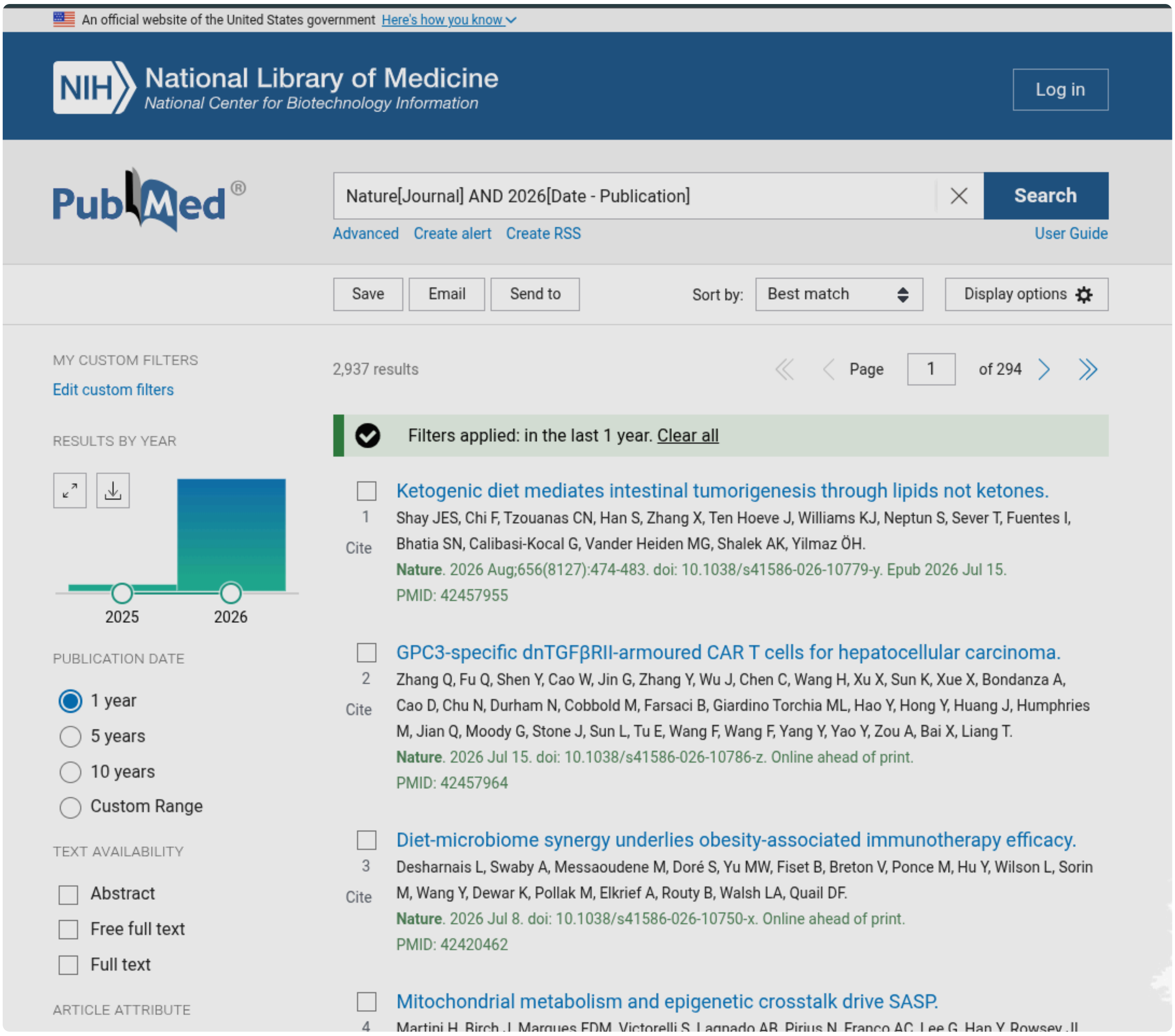
J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.yjmcc.2009.03.006. Epub 2009 Mar 19.

PMID: 19303419 Review.

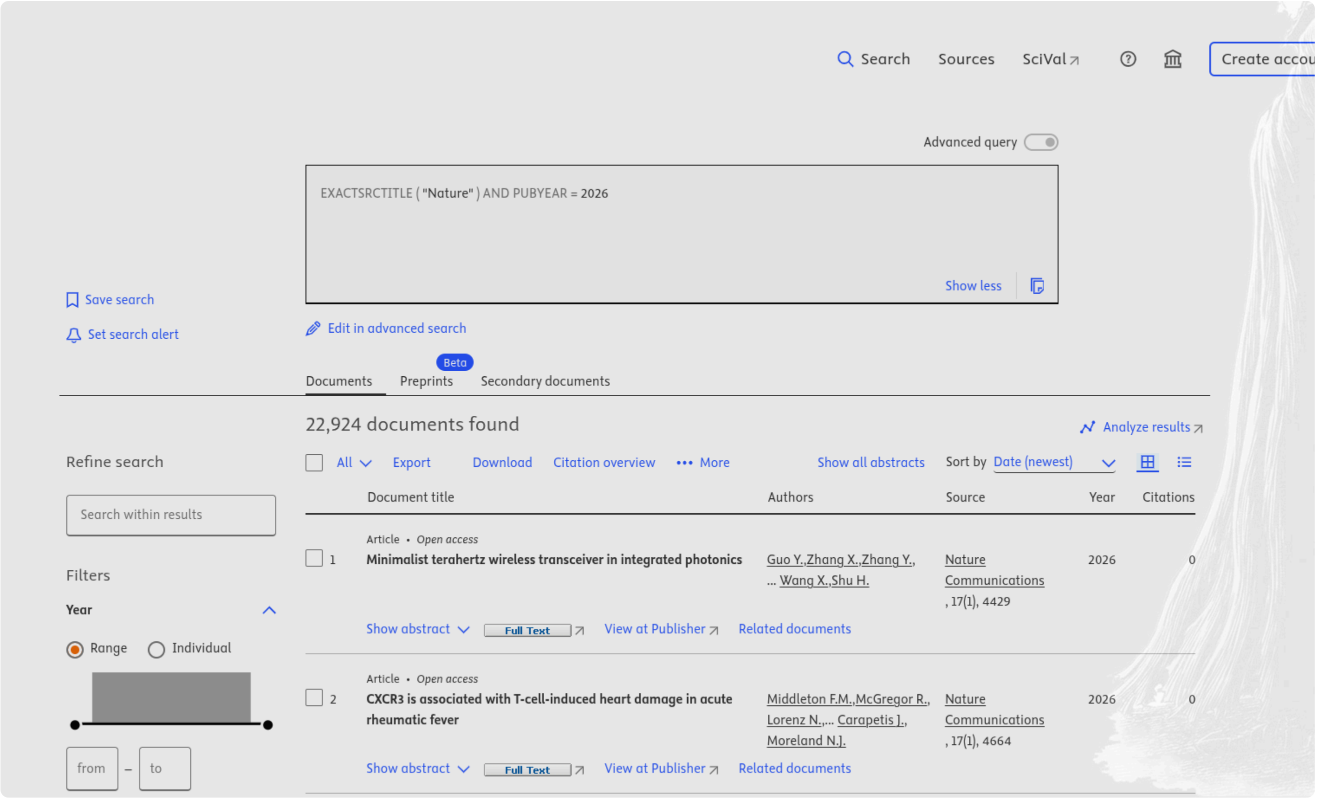
See all similar articles

Total 27 similar articles are listed here

Q.7

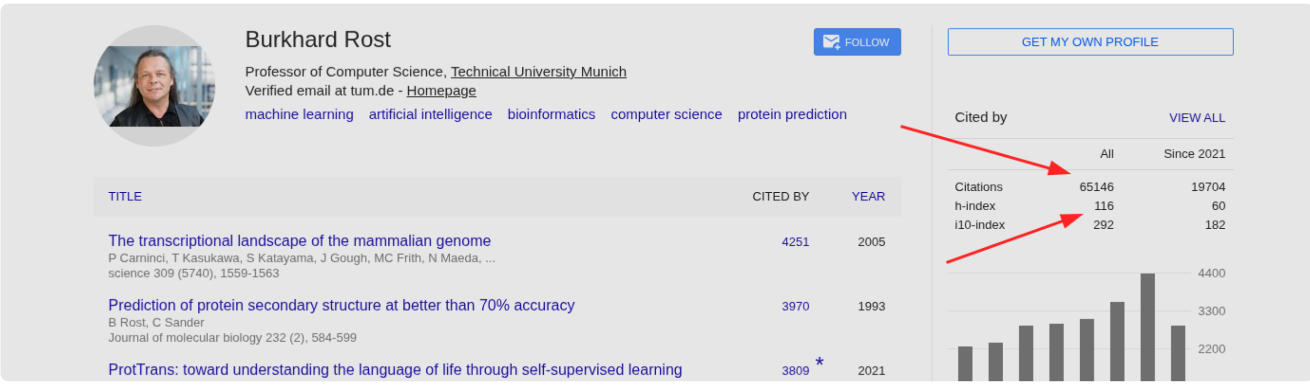


Pubmed shows 2937 articles



SCOPUS shows 22,924 articles

Q.8

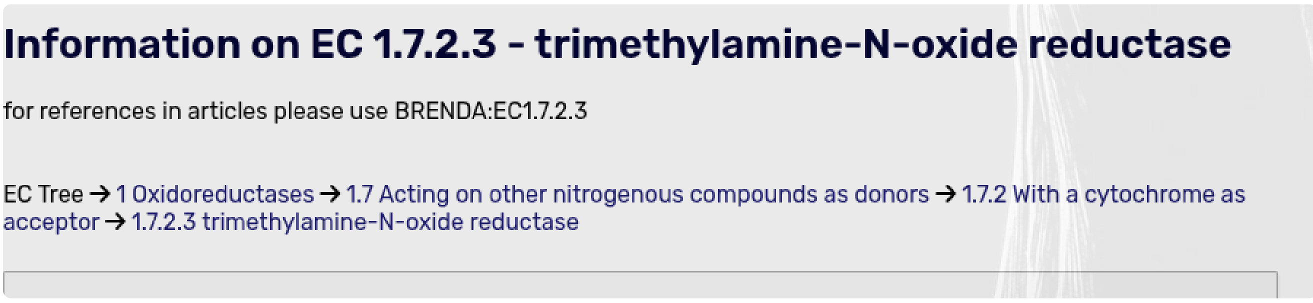


Citations = 65146

h-index = 116 (highest i have seen for any researcher till now!)

Q.9

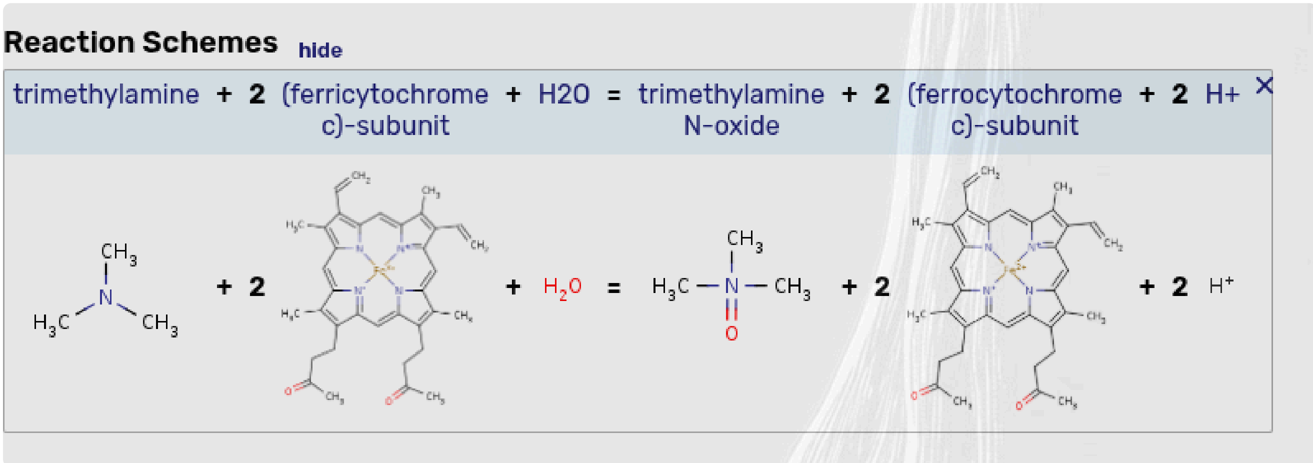
link to brenda



this shows the enzyme class hierarchy

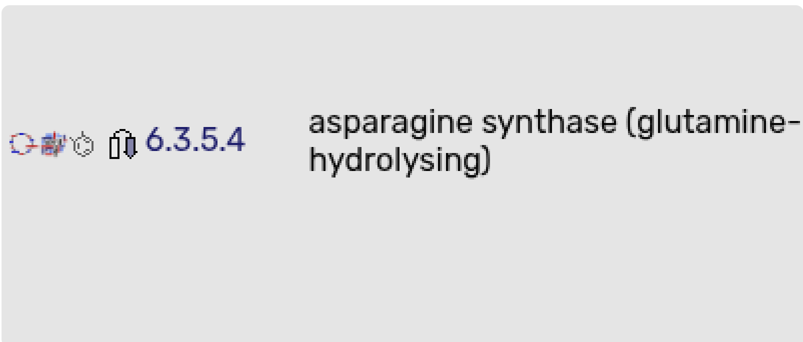
- **Accepted Name:** Trimethylamine-N-oxide reductase (cytochrome c), also commonly known as TMAO reductase.
- **Enzyme Class Hierarchy:**
  - **EC 1 (Class):** Oxidoreductases (enzymes that catalyze oxidation-reduction reactions).
  - **EC 1.7 (Subclass):** Acting on other nitrogenous compounds as donors.
  - **EC 1.7.2 (Sub-subclass):** With a cytochrome as an acceptor.
- **Function / Catalytic Activity:** This microbial enzyme catalyzes the chemical reduction of trimethylamine N-oxide (TMAO) into trimethylamine (TMA) as part of the electron transport chain.

reaction



Q.10

finding the EC number on brenda

- 

now searching it on ebi(throntron) gives us the enzyme https://www.ebi.ac.uk/thornton-srv/m-csa/entry/302/

PDB

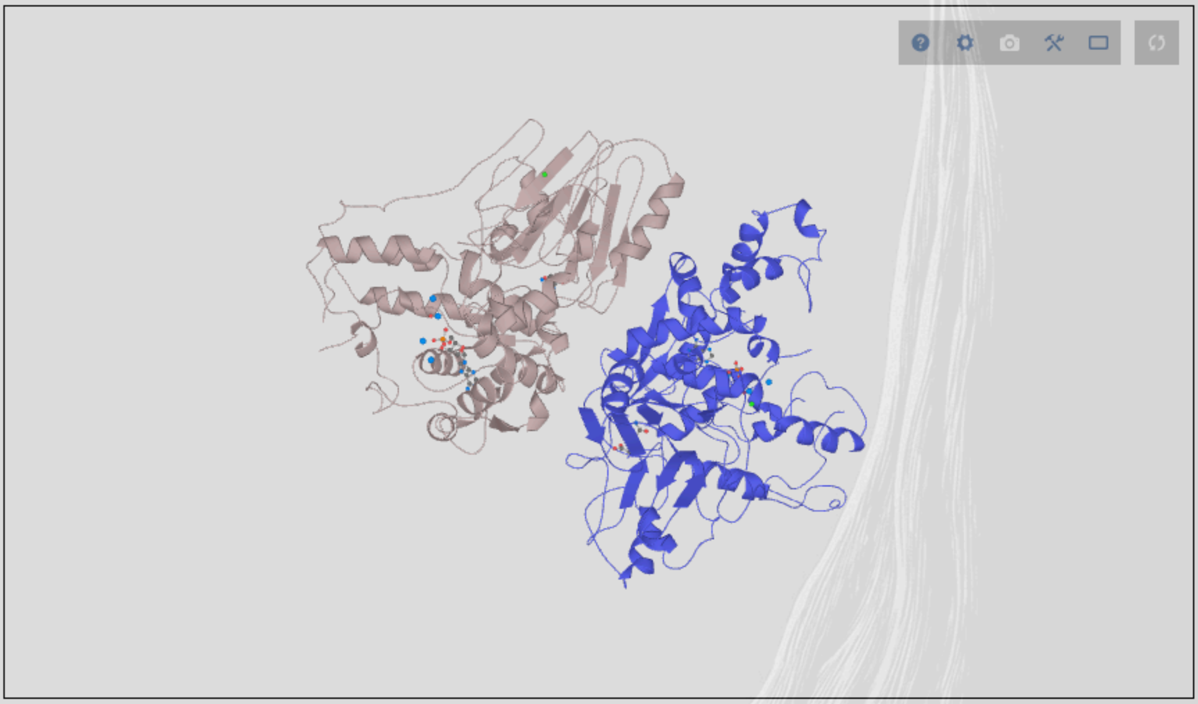
1ct9 - CRYSTAL STRUCTURE OF ASPARAGINE SYNTHETASE B FROM ESCHERICHIA COLI (2.0 Å)

Catalytic CATH Domains

3.60.20.10 3.40.50.620 (see all for 1ct9)

Cofactors

Magnesium(2+) (1) Metal MACiE



Show Active Center

and below it shows the catalytic residues and their roles

Catalytic Residues Roles

UniProt	PDB* (1ct9)		
Cys2 (N-term)	Ala1A (N-term)	Acts as a general acid/base to activate the cysteine nucleophile.	proton acceptor, proton donor
Leu51 (main-C)	Leu50A (main-C)	Helps stabilise the reactive intermediates formed.	hydrogen bond acceptor, electrostatic stabiliser
Thr322, Arg325	Thr321A, Arg324A	Bind and stabilise the phosphate groups of the ATP and reactive intermediates formed.	hydrogen bond donor, electrostatic stabiliser
Cys2	Ala1A	Acts as a catalytic nucleophile in the glutaminase domain reaction.	covalently attached, hydrogen bond acceptor, nucleofuge, nucleophile, proton acceptor, proton donor
Gly76 (main-N), Asn75	Gly75A (main-N), Asn74A	Forms the oxyanion hole that stabilises the reactive intermediates and transition states formed.	hydrogen bond donor, electrostatic stabiliser

so answer = Cys 1, Cys 2, Gly76, Leu 51, Thr 322, and Arg 325.

Q.11

Search for as complete name

Homo sapiens

Taxonomy ID: 9606 (for references in articles please use ncbitaxon:9606)

current name

Homo sapiens Linnaeus, 1758

Genbank common name: **human**
NCBI BLAST name: **primates**
Rank: **species**
Genetic code: Translation table 1 (Standard)
Mitochondrial genetic code: Translation table 2 (Vertebrate Mitochondrial)

Lineage*(full)*
cellular organisms; Eukaryota; Opisthokonta; Metazoa; Eumetazoa; Bilateria; Deuterostomia; Chordata; Craniata; Vertebrata;
Gnathostomata; Teleostomi; Euteleostomi; Sarcopterygii;

just listing the data below now

Organism	Scientific Name	Taxonomy ID	Number of Chromosomes (2n)
Human	*Homo sapiens*	9606	46
Cat	*Felis catus*	9685	38
Dog	*Canis lupus familiaris*	9615	78
Domestic guinea pig	*Cavia porcellus*	10141	64
Thale cress	*Arabidopsis thaliana*	3702	10

Q.12

The **Entrez Programming Utilities (E-utilities)** are a suite of server-side programs that serve as the public API for the NCBI Entrez system. They allow users to programmatically search, link, and retrieve data from across all NCBI Entrez databases (including PubMed, Nucleotide, Protein, and Gene) without needing to use the web browser interface.

example syntax

```
esearch.fcgi?db=<database>&term=<query>
```

example

- `[https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science[journal]+AND+breast+cancer+AND+2008[pdat]](https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science%5bjournal%5d+AND+breast+cancer+AND+2008%5bpdat%5d)`

to fetch a fasta format `[https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=nuccore&id=34577062,24475906&rettype=fasta&retmode=text]`

syntax is

- `db=nuccore`: Specifies the database (e.g., `nuccore` for nucleotide, or `protein`).
- `id=AY330867`: The unique identifier or accession number of the record.
- `rettype=fasta`: Tells the server you want the specific FASTA layout.
- `retmode=text`: Tells the server to return plain text rather than XML.

Q.13

a. Protein properties

1. **UniProtKB (Universal Protein Resource):** The central hub for the collection of functional information on proteins, with accurate, consistent, and rich annotation of protein properties.
2. **Pfam (or InterPro):** A large collection of protein families, represented by multiple sequence alignments and hidden Markov models (HMMs), used to determine protein domain properties.
   **b. Small molecules (Structure related)*
3. **PubChem:** A massive database maintained by NCBI that provides information on the chemical structures, physical properties, and biological activities of small molecules
4. **ChEBI (Chemical Entities of Biological Interest):** A freely available dictionary of molecular entities focused on strictly "small" chemical compounds and their structural properties.
   c. Cancer gene databases

5. **COSMIC (Catalogue of Somatic Mutations in Cancer):** The world's largest and most comprehensive resource for exploring the impact of somatic mutations in human cancer.
6. **cBioPortal for Cancer Genomics:** An open-access resource for interactive exploration of multidimensional cancer genomics data sets.